Restaurant Sales Prediction

Case study-
A restaurant wants to predict daily sales **based on factors like number of customers, day of the week, promotions, and weather conditions.**
Perform the following:
i) Build a suitable ML model
ii) Identify input and output variables
iii) Explain preprocessing steps required

Answer-

i) Build a Suitable Machine Learning Model (3–4 Marks)

Since **sales is a continuous value**, this is a **Regression problem**.

☞ Suitable Models:

- **Linear Regression** (simple and commonly used)
- Decision Tree Regressor (alternative)

Python Implementation (Linear Regression)

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Sample dataset
data = {
    'customers': [50, 60, 80, 100, 120],
    'day': [1, 2, 3, 4, 5],  # encoded (Mon=1,...)
    'promotion': [0, 1, 0, 1, 1],
    'weather': [1, 2, 1, 3, 2],  # encoded
    'sales': [5000, 6000, 7500, 9000, 11000]
}

df = pd.DataFrame(data)

# Input and output
X = df[['customers', 'day', 'promotion', 'weather']]
y = df['sales']

# Split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Model
model = LinearRegression()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)
print(y_pred)
```

ii) Identify Input and Output Variables (2–3 Marks)

Independent Variables (Input – X):

- Number of Customers
- Day of the Week
- Promotion (Yes/No)
- Weather Condition

Dependent Variable (Output – Y):

- Daily Sales

These factors influence restaurant revenue.

iii) Data Preprocessing Steps (Detailed – 4–5 Marks)

Data preprocessing is necessary because real-world restaurant data may be **incomplete, inconsistent, and unstructured**.

1. Handling Missing Values

- Remove or fill missing data
- Use mean/median for numerical values

Example: Missing number of customers → replace with average

2. Encoding Categorical Data

- Convert non-numeric data into numeric form

Example:

- Day (Mon, Tue) → numerical encoding
- Weather (Sunny, Rainy) → One-hot encoding

3. Feature Scaling

- Normalize values like customers and sales

Helps improve model performance and accuracy

4. Removing Outliers

- Identify abnormal values

Example: Extremely high sales on a festival day may distort model

5. Train-Test Split

- Split data into training (80%) and testing (20%)

Ensures model is tested on unseen data

Conclusion

- ✓ Regression model is suitable for predicting sales
- ✓ Proper preprocessing improves prediction accuracy
- ✓ Model helps restaurant in **planning inventory and promotions**